## (a) Constrained SGS-flux decomposition

$$q_{\mathrm{SGS}} = q_d + q_b$$

**Dissipative component**

$q_d = -\nu_d A_\theta \nabla\bar{\omega}, \quad A_\theta > 0$

$\Pi_d = -q_d \cdot \nabla\bar{\omega} \geq 0$ pointwise

**Bounded signed correction**

$q_b$: sign-indefinite local correction

Allows local reverse SGS enstrophy transfer

The pointwise guarantee belongs to the dissipative component; it is not a global stability theorem for the complete closure.

## (b–d) What information enters each evaluation mode?

| Mode | Reference / observation access | State update | Reported quantity |
|---|---|---|---|
| **A-priori reference state** | Reference state at every time | Closure evaluated on each reference state | SGS flux, source, transfer |
| **Free-running continuation** | Reference once at t = 0.1 s | Solver + closure; model state evolves | Trajectory error over 0.1–5 s |
| **Observation-assisted reconstruction** | 400 fixed sites (4%); ω, u, v every 0.1 s | Solver + fixed nudging; sparse data continue | Reconstruction over 0.1–10 s |

**Free-running after t = 0.1 s: no sensors, nudging, teacher forcing, reference reset or adaptation.**